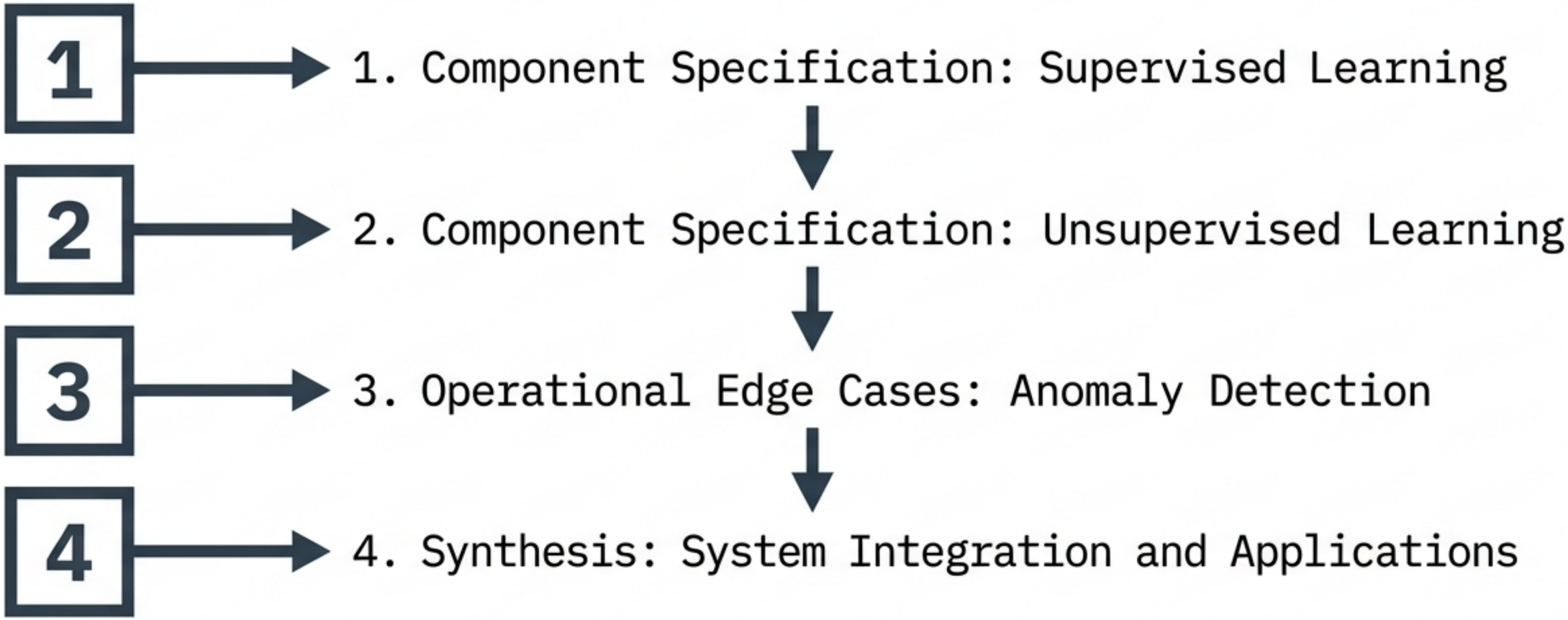
Module 4, Lesson 1

AI 101 for Mechanical Engineers

Learning with and without labels

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System Architecture Outline



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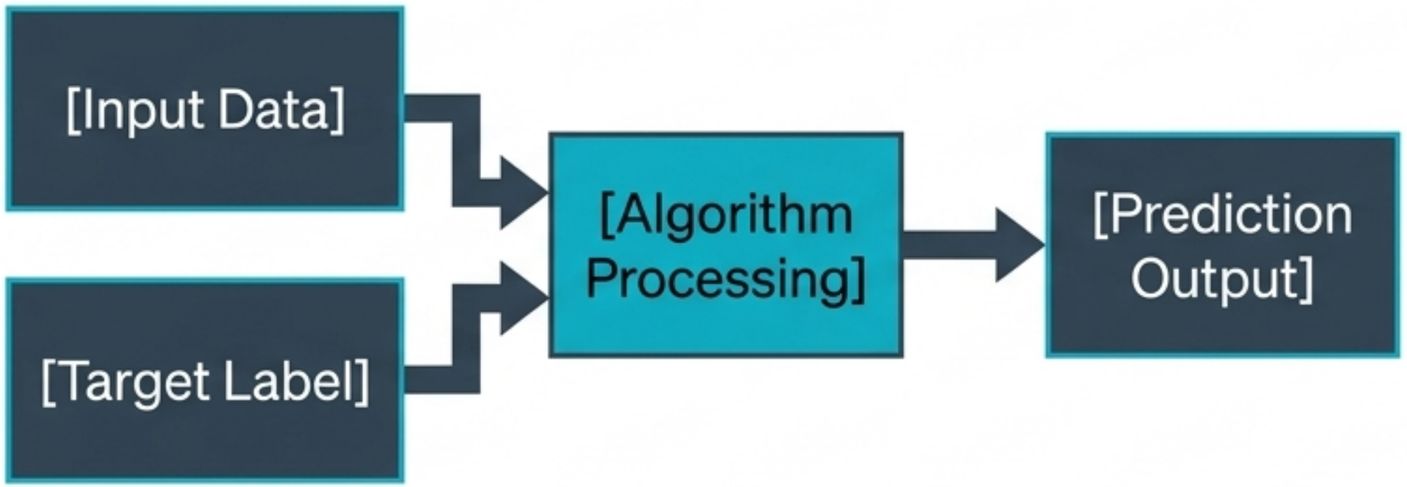
Section 01

Supervised Learning Specifications

Defining predictive paradigms driven by paired input-target data.

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Supervised Architecture

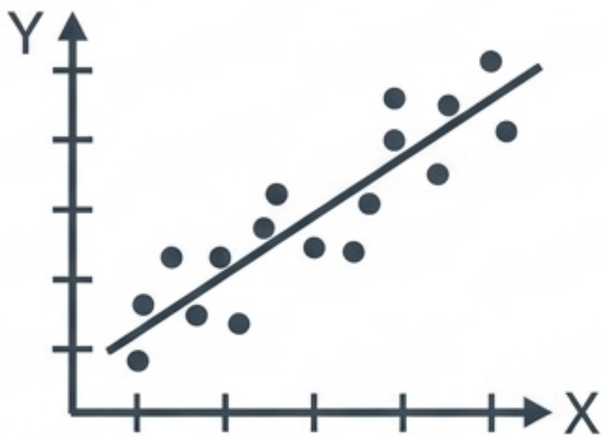


- Requires strictly paired input-target operational data.
- Model maps input features to known analytical targets.
- Target variable strictly dictates the specific learning task.
- Example: Predict power using known power measurements.

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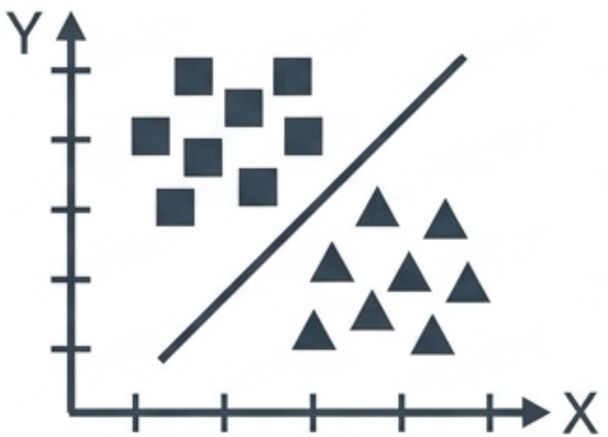
Output Topologies

Regression



- Predicts a continuous numerical output value.
- Examples: Tool wear (w) in [mm].
- Examples: Pressure drop in [Pa].

Classification



- Predicts a discrete category or class probability.
- Example: Differentiate intact components from cracked components.

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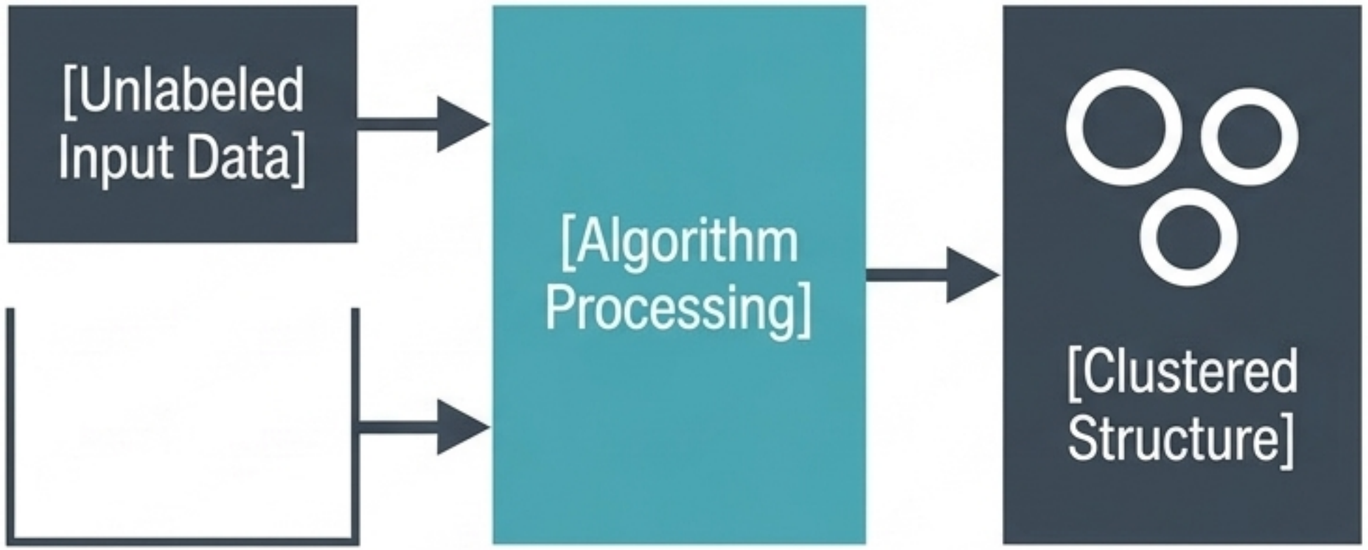
Section 02

Unsupervised Learning Specifications

Discovering underlying structural patterns without supplied target labels.

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Unsupervised Architecture



- Operates entirely without task-specific target labels.
- Algorithms identify underlying structural data similarities.
- Clustering groups examples by defined similarity criteria.
- Example: Group operational cycles or daily load profiles.
- Groups reflect machine regimes, not automatically mechanical health.

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Section 03

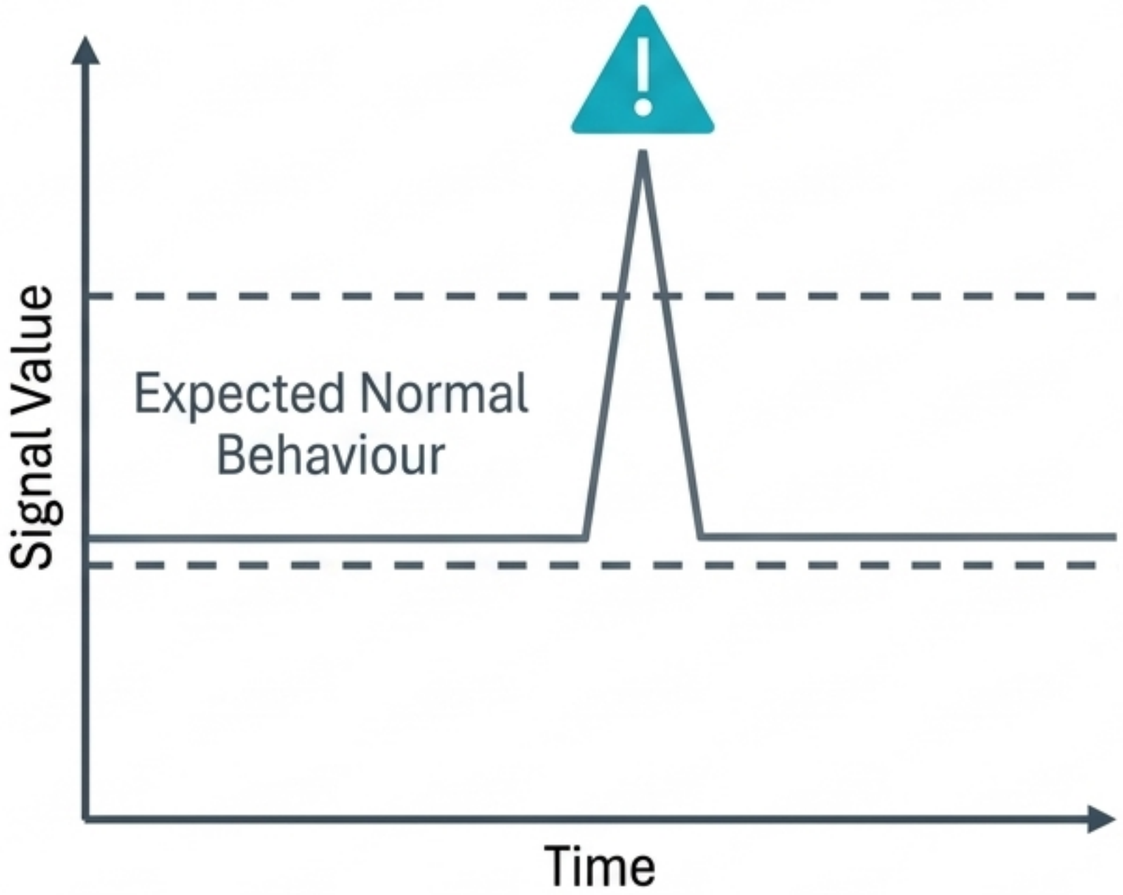
Operational Edge Cases: Anomaly Detection

Identifying and investigating deviations from established baseline behaviour.



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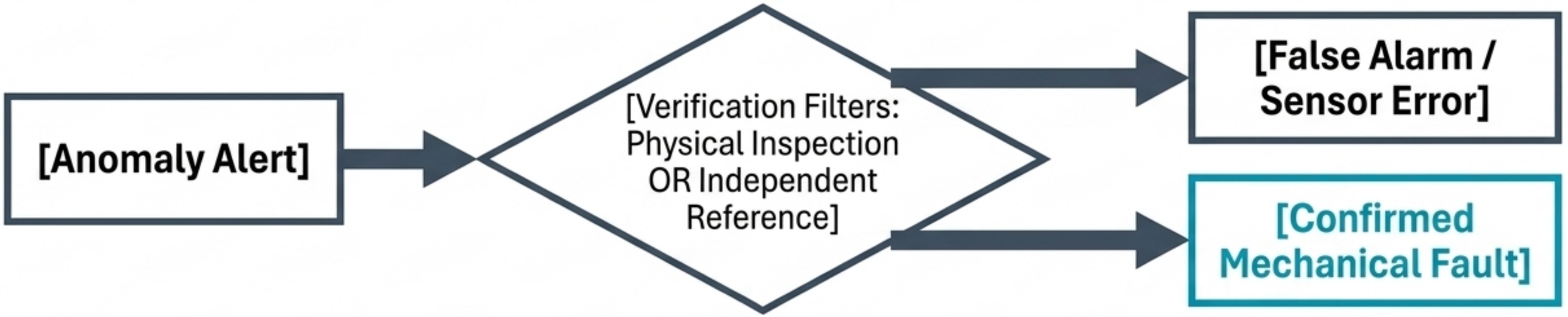
Defining System Anomalies



- Flags data deviating from expected baseline operational behaviour.
- Alerts mandate physical investigation, not proof of actual fault.
- Causes include new external mechanical structural loads.
- Causes include intentionally altered operational control settings.
- Causes include faulty internal sensing mechanisms or drift.

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Anomaly Verification Logic



- Unusual statistical patterns demand external physical structural validation.
- Physical inspections isolate actual mechanical equipment damage.
- Independent technical references establish verified true fault conditions.
- Always rule out new operating regimes or sensor transmission failures.

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Section 04

System Integration Synthesis

Mapping artificial intelligence paradigms to specific mechanical maintenance tasks.

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Maintenance Application Matrix

Learning Setup Diagnostic Table		
Maintenance Application	**Input-Target Availability**	**Learning Paradigm**
Wear Prediction	Supplied input-target pairs	Supervised Regression
Grouping Regimes	No target for task	Unsupervised Clustering
Masked Reconstruction	Target from original signal	Self-supervised Objective
Simulated Controller	Reward signals post-action	Reinforcement Learning

- Algorithm architecture depends **entirely upon available** operational data feedback.
- Supervised paradigms require **measured historical** component **targets**.
- Unsupervised paradigms discover inherent operational system groupings.
- Reinforcement paradigms optimize policies via simulated procedural interactions.



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Key Points Summary



Section 1: Supervised models map inputs to known analytical targets.



Section 2: Unsupervised models discover internal structures without labels.



Section 3: Anomaly alerts demand structural physical verification procedures.



Section 4: Available mechanical data topology strictly dictates the learning paradigm.

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